

Safety Data Sheet

1. Chemical product and company identification

Product name : Sodium hydroxide

Company information

Name of manufacturer : KANTO CHEMICAL CO., INC.
 Address : 2-1, Nihonbashi, Muromachi 2-Chome, Chuo-Ku, Tokyo, 103-0022, JP
 Name of section : Business Administration Department, Reagent Division
 Telephone number : +81-3-6214-1090
 Facsimile number : +81-3-3241-1047
 Mail address : BC32@kanto.co.jp
 Reference No : 37184
 Product numbers applied by the SDS : 37003, 37184, 37186, 37943, 37944, 37945, 37990

2. Hazards identification

GHS classification

Health hazards	Skin corrosion/irritation	Category 1B
	Serious eye damage/eye irritation	Category 1
	Specific target organ toxicity (single exposure)	Category 1 (respiratory organs)
Environmental hazards	Aquatic acute	Category 3

Hazard pictograms



Signal word : Danger

Hazard statements : Causes severe skin burns and eye damage
 Causes damage to organs (respiratory organs)
 Harmful to aquatic life

Precautionary statements

Prevention : Do not breathe dust.
 Wash hands, forearms and face thoroughly after handling.
 Do not eat, drink or smoke when using this product.
 Avoid release to the environment.
 Wear protective gloves/protective clothing/eye protection/face protection.

Response : IF SWALLOWED: Rinse mouth. Do not induce vomiting.
 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water .
 IF INHALED: Remove person to fresh air and keep comfortable for breathing
 IF IN EYES: Rinse cautiously with water for several minutes.
 Remove contact lenses, if present and easy to do. Continue rinsing.



IF exposed or concerned: Call a POISON CENTER or doctor.
Immediately call a POISON CENTER or doctor.

Storage : Store locked up.

Disposal : Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

3. Composition/information on ingredients

Distinction of substance or mixture : Substance

Chemical name	Concentration (%)	Formula	TSCA	EC-No.	CAS RN
Sodium hydroxide	≥ 95	NaOH	Listed	215-185-5	1310-73-2

4. First aid measures

First aid measures

First-aid measures after inhalation : Remove the victim to fresh air, and make him blow his nose and gargle.

First-aid measures after skin contact : Wash the affected areas under running water, get medical treatment as soon as possible.

First-aid measures after eye contact : Wash the affected areas under running water for at least 15 minutes. Get medical treatment.

First-aid measures after ingestion : Rinse mouth with water. Give the victim one or two glasses of water or milk. Do not induce vomiting. Get medical treatment as soon as possible.

Personal Protection in First Aid and Measures : Rescuers should wear proper protective equipment like rubber gloves, goggles.

Most Important Symptoms/Effects

Symptoms/effects : Inhalation brings on burning sensation of throat, throat pain, cough, breathlessness, and these symptoms may delay. Skin contact causes redness, pain, severe skin burns, and blisters. Eye contact causes redness, pain, and blurred vision.

5. Fire fighting measures

Suitable extinguishing media : This product is noncombustible.

Unsuitable extinguishing media : None

Firefighting instructions : Move containers from fire area if it can be done without risk, if not possible, apply water from a safe distance to cool and protect surrounding area.

Personal protection (Emergency response) : Firefighters should wear protective equipment.

6. Accidental release measures

Personal Precautions, Protective Equipment and Emergency Procedures

General measures : Wear proper protective equipment and avoid contact with skin and inhalation of dust. Conduct operations from upwind and evacuate



people downwind.

Environmental precautions

Environmental precautions : Attention should be given to avoid discharge of spilled product into rivers and resulting environmental damage. When diluting spill with large amounts of water, discharge of untreated wastewater into the environment must be avoided.

Methods and Equipment for Containment and Cleaning up

For containment : Sweep up in a chemical waste container. Flush contaminated area with copious amounts of water.

7. Handling and storage**Handling**

Technical measures : Wear appropriate protective equipment to avoid contact with skin or inhalation of dust.

Precautions for safe handling : Avoid formation of dust and aerosols.

Storage

Storage conditions : As the chemical is deliquescent, keep the bottle tightly closed and store in a cool place.

Material used in packaging/containers : Polyethylene, polypropylene, fluorocarbon polymers.

8. Exposure controls / Personal protection equipment

ACGIH Ceiling	2 mg/m ³
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Appropriate engineering controls : Install a local ventilation system in case of dusty condition.

Protective equipment

Respiratory protection : If necessary, wear dust mask

Hand protection : Impervious protective gloves

Eye protection : Safety goggles

Skin and body protection : Protective clothing, protective boots

9. Physical and chemical properties

Physical state : Solid

Color : White

Odor : Odorless

pH : 14 (5% aqueous solution)

Melting point : 318 ° C

Freezing point : No data available

Boiling point : 1388 ° C

Flash point : No data available

Auto-ignition temperature : No data available

Decomposition temperature : No data available

Flammability (solid, gas) : Non flammable

Vapor pressure : No data available

Relative density : No data available



Density	: 2.1 g/cm ³ (20°C)
Relative gas density	: No data available
Solubility	: Organic solvents: Soluble in ethanol, glycerol, methanol. Water: 55 % (20°C)
Partition coefficient n-octanol/water (log Pow)	: No data available
Explosive limits (vol %)	: No data available
Viscosity, kinematic:	: No data available
Particle characteristics	: No data available

10. Stability and reactivity

Reactivity	: Hydroxides precipitate from aqueous solutions of many metal salts. The molten salt attacks gold and platinum and melts the silicate. Fats are saponified to produce the fatty acids sodium and glycerin.
Chemical stability	: Stable under normal conditions. Deliquescent. It absorbs moisture and carbon dioxide in the air to produce sodium carbonate.
Possibility of hazardous reactions	: Generate heat vigorously when contact with acids. It also generates heat when dissolved in water. The chemical corrodes aluminium, tin, zinc, chromium, and their alloys, releasing combustible hydrogen gas.
Conditions to avoid	: Light, heat, moisture.
Incompatible materials	: Acids, metals.
Hazardous decomposition products	: Sodium oxide, hydrogen.

11. Toxicological information

Acute toxicity (oral)	: Classification not possible Since acute toxicity data is only rabbit oral LD50=325mg/kg, and there is no data in rodents, the classification on acute oral toxicity is not possible.
Acute toxicity (dermal)	: Classification not possible
Acute toxicity (inhalation)	: No classification (gas) No classification (vapor) Classification not possible (dust, mist)
Skin corrosion/irritation	: Causes severe skin burns In a pig test using application of 2N (8%), 4N (16%) and 6N (24%) solutions on the abdominal region, gross blisters developed within 15 minutes of application and the 8 and 16% solutions produced severe necrosis in all epidermal layers. The 24% concentration produced numerous and severe blisters with necrosis extending deeper into the subcutaneous tissue. Additionally, there is a report that severe necrosis occurred after application of a 5% solution to the skin of rabbits for 4-hour. Based on these data, the substance was classified into category 1B. As relevant information, the pH is 12 (0.05%). For humans, 0.5 - 4% solutions were irritating to the skin, and in skin irritation tests with a 0.5% solution, 55% and 61% of the volunteers showed positive skin irritation reactions.
Serious eye damage/irritation	: Causes serious eye damage Based on a report that the corrosive concentration for rabbit eyes was 1.2% or higher than 2%, and a pH of 12 (0.05%), it was classified into category 1.
Respiratory sensitization	: Classification not possible



Skin sensitization	: No classification Male volunteers were exposed on their backs to concentrations of 0.063 - 1.0% of the substance. After 7 days the volunteers were challenged to a concentration of 0.125%. The irritant response correlated well with the concentration, but an increased response was not observed when the previously patch tested sites were rechallenged. Based on these results, sodium hydroxide has no skin sensitization potential.
Germ cell mutagenicity	: No classification In vivo mouse bone marrow micronuclei test is negative, and in vitro mutagenicity test (Ames test) is also negative.
Carcinogenicity	: Classification not possible
Reproductive toxicity	: Classification not possible
STOT-single exposure	: Causes damage to organs (respiratory organs) Based on the description that acute inhalation exposure of dust or mist causes mucous membrane irritation followed by cough and breathing difficulty, and more intensive exposure may cause pulmonary edema or shock, it was classified into category 1(respiratory organs).
STOT-repeated exposure	: Classification not possible
Aspiration hazard	: Classification not possible

12. Ecological information

Ecotoxicity

Aquatic acute	: Harmful to aquatic life Cenodaphnia quadrangular LC50=40mg/L/48h
Aquatic chronic	: No classification Classified into "No classification" since the toxicity is mitigated in environmental water by buffer action though it is considered to be a factor of toxicity that water solution becomes a strong base.

Persistence and degradability

No additional information available

Bioaccumulative potential

No additional information available

Mobility in soil

No additional information available

Hazardous to the ozone layer

Ozone	: Classification not possible
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13. Disposal considerations

Ecology - waste materials	: Neutralization method : Dissolve in water and flush in a drain after neutralizing with diluted acids. Or consult approved disposal companies.
Contaminated container and packaging	: In case of disposal of empty bottles, dispose bottles after removing the content thoroughly.



14. Transport information

International Regulations

Transport by sea(IMDG)

UN-No. (IMDG) : 1823
Proper Shipping Name (IMDG) : SODIUM HYDROXIDE, SOLID
Packing group (IMDG) : II
Transport hazard class(es) : 8

(IMDG)

Air transport(IATA)

UN-No. (IATA) : 1823
Proper Shipping Name (IATA) : Sodium hydroxide, solid
Packing group (IATA) : II
Transport hazard class(es) : 8

(IATA)

Marine pollutant : Not applicable

MFAG-No : 154

15. Regulatory information

Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.

16. Other information

Data sources : Handbook of dangerous and hazardous chemicals, Japan Industrial Safety & Health Association. (2000-2001).
Handbook of Dangerous Substances Springer-Verlag Tokyo(1991).
Handbook of 17221 Chemical Products, The Chemical Daily Co. (2021).
Handbook of Poisonous and Deleterious substances, revised and enlarged edition, Yakumu Kohosa(2000).
NITE Chemical Risk Information Platform (NITE-CHRIP), National Institute of Technology and Evaluation.

The information contained herein is based on several references and the present state of our knowledge. However the SDS does not always cover all information about the product, handle the product carefully. The information is intended to ordinary usage, in case of particular handlings, conduct appropriate safety measurements. The information herein is only provision of information, and it does not represent a guarantee the properties of the product. The Safety Data Sheet(SDS) is prepared based on JIS Z7253.

